AYSHA IRAM

Machine Learning | Computer Vision | NLP | AI & Data Science
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PROFESSIONAL SUMMARY

AI & Data Science undergraduate (B.E. 2027, CGPA 8.5) with hands-on machine learning across computer vision, NLP and risk modelling. Fine-tuned and compared vision models (ViT, ResNet18) with attention-map interpretability, built dense and sparse retrieval pipelines for grounded question answering, and contribute to an explainable, ablation-tested transaction-risk model selected by the college for Smart India Hackathon 2026. Strong in Python, familiar with Java and C, and experienced in taking models from experiment to working system.

EDUCATION

B.E. — AI & Data Science, Don Bosco Institute of Technology, Bengaluru (VTU) | CGPA: 8.5 2023 – 2027

12th (PCMC), Scholars PU College, Bengaluru | 94.5% 2022

TECHNICAL SKILLS

Languages: Python (primary), Java, C, JavaScript, SQL
ML & Deep Learning: PyTorch, Hugging Face Transformers & Datasets, TensorFlow / Keras, Scikit-learn, XGBoost
Computer Vision: Vision Transformers, CNNs (ResNet), transfer learning, image classification, attention-map visualization
NLP & LLMs: Retrieval-augmented generation, Sentence-Transformers, FAISS, TF-IDF retrieval, local LLMs (Llama 3), multi-agent systems (CrewAI)
Data & Evaluation: Pandas, NumPy, DuckDB | precision / recall / F1, ablation studies, probability calibration
Tools: Git/GitHub, Jupyter, FastAPI, MongoDB, Node.js, Power BI, Tableau

EXPERIENCE

Software Developer Trainee — Commonwealth Bank of Australia (CBA) Jul 2026 – Jan 2027

- Built a **Retail Banking & AML Transaction Monitoring System** (GitHub) with a configurable rule engine (large transaction, high frequency, structuring) and explainable 0–100 risk scoring for every flagged transaction.
- Backed it with **112 automated unit and integration tests**, idempotent transfers with rollback, and full audit logging.

AI with Python Intern — Languify Aug 2024 – Sep 2024

- Built and evaluated applied ML models with Keras, Pandas and Vision Transformers; wrote preprocessing and experiment scripts and compared results across metrics to verify model quality.

PROJECTS

Trinetra — AI Bitcoin Transaction Monitoring | SIH 2026 | GitHub Python, XGBoost, DuckDB, FastAPI

- **Selected by the college to represent DBIT at Smart India Hackathon 2026** (Team Sentinels).
- Links network data (IP, ASN, timing) with blockchain data (wallets, transactions) across **500k observations**, resolves addresses into entities, and ranks them with an XGBoost model into an explainable lead queue with per-alert SHAP reasons.
- Every feature family **ablation-tested** for its contribution; calibrated probabilities evaluated with PR-AUC and precision@k against a rules baseline; fully offline, reproducible pipeline with 171 automated tests.

Real-Time Behavioural Surveillance | Final-Year Project YOLOv8, MediaPipe, LSTM, Spark

- Building a distributed computer-vision pipeline: Kafka ingests multi-camera CCTV frames, Spark Structured Streaming runs **YOLOv8n** object detection and **MediaPipe BlazePose** (33 keypoints) per frame, and a 2-layer **LSTM** classifies 30-frame pose sequences.
- Designed a scalability benchmark measuring latency, throughput and accuracy from 1 to 10 camera feeds, distributed vs single-node.

Plant Disease Classification — Vision Transformer vs CNN | GitHub PyTorch, Hugging Face

- Fine-tuned ViT-Base/16 (86M parameters) with the Hugging Face Trainer and ResNet18 (11M) via transfer learning on the Beans dataset (1,295 images, 3 classes); ResNet18 reached **97.7% test accuracy**.
- Implemented CLS-token **attention-map visualization** to interpret ViT predictions; evaluated with per-class precision, recall, F1 and confusion matrices.

Educational Atlas — Retrieval-Augmented QA over Textbooks | GitHub FAISS, Sentence-Transformers, Llama 3

- Built a RAG pipeline over 4 open textbooks (**2,300+ chunks**): PDF parsing and chunking, dense retrieval (MiniLM embeddings + FAISS) alongside a TF-IDF sparse baseline, and grounded answer generation with a local Llama 3 model.

ACHIEVEMENTS & CERTIFICATIONS

- **Smart India Hackathon: selected by the college two years running** — MindCare (SIH 2025) and Trinetra (SIH 2026).
- Python and AI — Amazon Future Engineer (Navgurukul) | Open Source Program — Google Developer Student Club
- Full Stack Web Development with AI Tools — Edunet Foundation, EY & AICTE | Power BI & Tableau Dashboards